

DIGITAL FORENSICS INVESTIGATION SUMMARY REPORT

EFCC Case Investigation • Digital Evidence Analysis of cartel.img

Case Title:	Examination of cartel.img	Date of Report:	3 August 2026
Module:	Digital Forensics & Incident Response	Examiner:	Okirie Chukwumeniem Faith
Subject Evidence:	cartel.img (USB Forensic Image)	Investigating Agency:	EFCC Forensic Division

1. EXECUTIVE SUMMARY & OBJECTIVES

At 09:15 AM on 2 August 2026, officers of the Economic and Financial Crimes Commission (EFCC) executed a search warrant during an active investigation, seizing a USB storage device from a suspect's residence. The suspect claimed that *"The USB only contains personal photographs."* A forensic bit-stream image named `cartel.img` was provided for analysis. The objective of this forensic examination is to identify, preserve, extract, and document all available digital evidence to verify or disprove the suspect's statement while maintaining strict chain-of-custody integrity.

2. EVIDENCE INTEGRITY & INITIAL TRIAGE

Prior to analysis, cryptographic baseline hashes were calculated for `cartel.img` to verify evidence preservation. Initial volume triage was conducted using Sleuth Kit tools to determine file system schema.

PARAMETER / METRIC	TECHNICAL SPECIFICATION / CRYPTOGRAPHIC VALUE
MD5 Hash	80348c58eec4c328ef1f7709adc56a54
SHA-256 Hash	ce550124200a997c61b413941cbef4df9619a2f96579674952294a176a32be65
Image Format & Size	Raw Format (raw) • Size: 259,506,176 bytes (259 MB / 247 MiB)
File System Architecture	FAT16 File System • 1 Partition • Sector Size: 512 bytes

3. ARTIFACT DISCOVERY & CARVED EVIDENCE ANALYSIS

Data carving using PhotoRec 7.1 and Foremost recovered **132 files** from unallocated cluster space. Low-level file system extraction using Sleuth Kit utilities (`fls`, `istat`, `icat`) recovered active directory assets.

Key Finding — Refutation of Suspect Assertion

Forensic carving recovered a deleted Microsoft Word document (`00335017.doc`) containing **1,022 words** of text, authored by **"NOWAY MAN"**. The recovery of this substantial text document conclusively refutes the suspect's claim that the USB contained only personal photographs.

ORIGINAL FILE	RECOVERED NAME	TYPE	RECOVERY METHOD	FORENSIC SIGNIFICANCE
She_died_in...doc	00335017.doc	MS Word	PhotoRec / Foremost	1,022 words; Author: NOWAY MAN; refutes photo-only claim.
gumbo1.txt	gumbo1.txt	Text	Sleuth Kit (Inode 4)	Shrimp & Tasso Gumbo recipe text; active inode allocation.
gumbo2.txt	gumbo2.txt	Text	Sleuth Kit (Inode 6)	Shrimp & Andouille Sausage recipe text (Bon Appetit 1992).
Unallocated	f0104057.jpg	JPEG	PhotoRec Carving	Recovered black-and-white graphic asset (Rhino drawing).
Unallocated	f0106865.gif	GIF	PhotoRec Carving	Recovered photographic image asset (Alligator photo).

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4. TIMELINE RECONSTRUCTION & ACTIVITY ANALYSIS

System timelines were reconstructed by parsing directory bodyfiles generated with fls through mactime. The resulting temporal record highlights user activity and metadata updates.

TIMESTAMP (UTC/LOCAL)	TARGET ARTIFACT	MACB FLAG	FORENSIC ACTIVITY & CONTEXT
Fri Apr 30 2004 00:00:00	/gumbo1.txt /gumbo2.txt	. a . . (Access)	System read access recorded for both text recipe artifacts in FAT root.
Fri Apr 30 2004 18:11:20	/gumbo1.txt (Inode 4)	m . . b (Mod/Attr)	File content modification and directory attribute metadata update.
Fri Apr 30 2004 18:11:24	/gumbo2.txt (Inode 6)	m . . b (Mod/Attr)	Secondary file content modification and metadata update logged.
Aug 02 2026 09:15 AM	Physical USB Device	Seizure Event	Evidence recovered by EFCC officers during search warrant execution.
Aug 03 2026	cartel.img	Analysis Event	Forensic hashing, triage, carving, and timeline analysis performed.

5. TOOLS & METHODOLOGY

The forensic workflow strictly adhered to NIST SP 800-86 standards for digital evidence handling:

- **Evidence Verification:** Computed MD5 and SHA-256 baseline hashes via standard Linux terminal utilities.
- **Volume & File System Triage:** Parsed FAT16 volume architecture using Sleuth Kit tools (mmls, img_stat).
- **Unallocated Space Carving:** Carved 132 files using PhotoRec 7.1 and verified redundantly via Foremost.
- **Inode Extraction & Verification:** Extracted file streams with icat and rendered documents using LibreOffice Writer.
- **Temporal Analysis:** Generated file system bodyfiles using fls and compiled timelines via mactime.

6. FORENSIC CONCLUSIONS & RECOMMENDATIONS

Conclusions: The forensic analysis of cartel.img was executed successfully without altering evidence integrity. The recovery of a 1,022-word Microsoft Word document (00335017.doc) authored by "NOWAY MAN", along with text files and image assets, conclusively invalidates the suspect's claim that the USB held only personal photographs.

Limitations & Recommendations: Physical drive serial numbers and specific deletion timestamps were not logged in image metadata. It is recommended to perform keyword indexing across all 132 carved files and cross-reference author metadata ("NOWAY MAN") against seized secondary media.

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Role: Senior Technical Documentation Specialist
Signature: Okirie C. Faith
Date: 3 August 2026

Investigating Agency: EFCC Forensic Unit
Verification Status: Passed (Hashes Verified)
Evidence Image: cartel.img (FAT16)
Total Artifacts Carved: 132 Files